

Introduction to Using HPRC Computing Clusters

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Host Institution: Prairie View A&M University

Workshop Venue: University of Houston

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Agenda

- Introduction to computing systems platform (ACCESS)  Advancing Innovation
- Introduction to High Performance Research Computing (HPRC) 
- Performance comparison of Advanced GPUs and IPU
- Hands on AI training with A30 GPUs .

Introduction to ACCESS



- ACCESS (Advanced CyberInfrastructure for Computational and Emerging Sciences) is a program funded by the NSF, designed to help researchers and educators with using HPC resources and cyberinfrastructure. Its aim is to make these resources more accessible to researchers. We can use the nation's advanced computing systems and services **at no cost**.
- TAMU is a member of the ACCESS program. TAMU'S High Performance Research Computing (HPRC) provides access to its high-performance computing resources, such as the ACES cluster, & Launcher cluster to researchers and educators through the ACCESS platform. HPRC provides a wide range of advanced AI accelerators.



- Collaborative connection between PVAMU with TAMU.



Introduction to HPRC

High Performance Research Computing

A Resource for Research and Discovery



The HPRC group currently administers six HPC clusters.

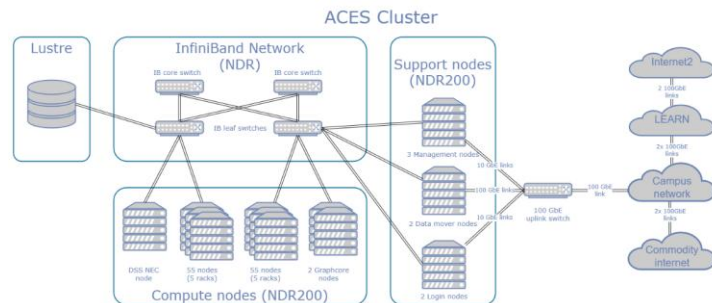
- ACES
- Launch
- FASTER
- Grace
- ViDaL
- Lonestar6

Introduction to ACES



ACES: A Dell x86 HPC Cluster

System Name:	ACES
Operating System:	Red Hat Enterprise 8
Total Compute Cores/Nodes:	11,888 cores 130 nodes
Compute Nodes:	110 Intel Sapphire Rapids Nodes 17 Intel Ice Lake Nodes 1 AMD Rome Graphcore Node with 16 Mk2 Colossus GC200 IPUs 1 Intel Ice Lake Graphcore Node with 16 Bow-2000 IPUs 1 Intel Cascade Lake Node with 8 NEC Vector Engine Type 20B-P cards
Composable Components	30 NVIDIA H100 GPUs 4 NVIDIA A30 GPUs 3 Bittware Agilex FPGAs 2 Intel D5005 FPGAs 2 NextSilicon coprocessors 48 Intel Optane SSDs 120 Intel GPU Max 1100
Interconnect:	NVIDIA Mellanox NDR200 InfiniBand (MPI and storage) Liquid PCIe Gen4 Fabrics (composability) Liquid PCIe Gen5 Fabrics (composability)



HPRC's ACES Cluster

ACES is a Dell cluster with rich accelerator testbed. 130 nodes. ACES cluster has a variety of HPC resources, Graphcore Intelligent Processing Unit (IPU), NVidia A100/H100 GPUs, Intel Max GPUs, and FPGA systems.

Introduction to FASTER

FASTER: A Dell x86 HPC Cluster



System Name:	FASTER
Host Name:	faster.hprc.tamu.edu
Operating System:	Rocky Linux 8
Total Compute Cores/Nodes:	11,520 cores 180 nodes
Compute Nodes:	180 64-core compute nodes, each with 256GB RAM
Composable GPUs:	200 T4 16GB GPUs 40 A100 40GB GPUs 8 A10 24GB GPUs 4 A30 24GB GPUs 8 A40 48GB GPUs
Interconnect:	Mellanox HDR100 InfiniBand (MPI and storage) Liquid PCIe Gen4 (GPU composability)
Peak Performance:	1.2 PFLOPS

HPRC's FASTER Cluster

- FASTER is a 184-node Dell cluster .
- FASTER composable cluster allow user to add up to 32 GPUs to a single computing node.
- A100 GPUs, A10 GPUs, A30 GPUs, A40 GPUs and T4 GPUs are distributed and composable via Liquid PCIe fabrics.

Introduction to Launch

Launch: A Dell x86 HPC Cluster

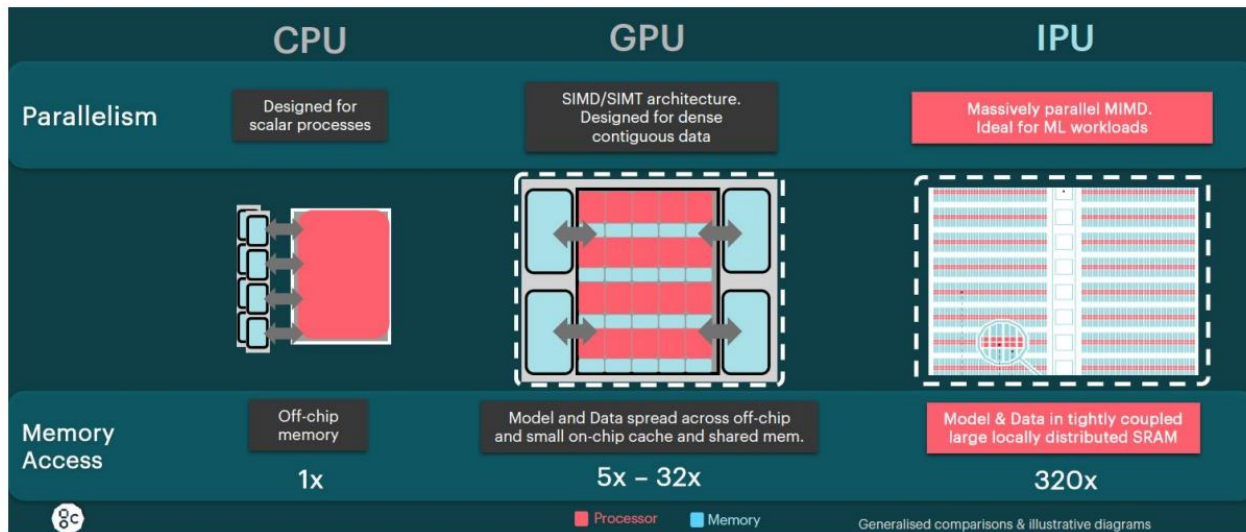
System Name:	Launch
Operating System:	Red Hat Enterprise 8
Total Compute Cores/Nodes:	8640 cores 45 nodes
Compute Nodes:	35 AMD Genoa CPU Nodes with 384 GB 10 AMD Genoa GPU nodes each with 2 NVIDIA A30s and 768 GB
Interconnect:	NVIDIA HDR100
Peak Performance:	436 TFLOPS
Global Disk:	2.2 PB (usable) via Dell storage server
File System:	NFS
Batch Facility:	Slurm by SchedMD
Location:	West Campus Data Center
Production Date:	Late 2023 or early 2024

Launch Cluster

- .Launch is a Dell Linux cluster with 45 compute nodes (8,640 cores) and 2 login nodes.
- 35 Genoa CPU Nodes.10 NVIDIA A30 GPU nodes.

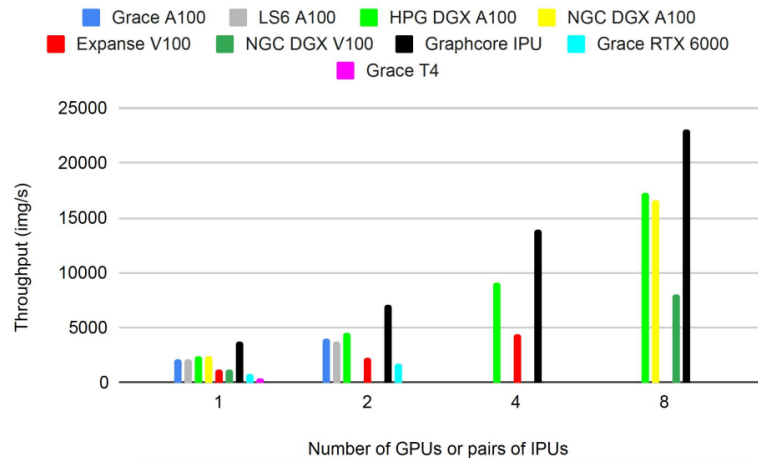
More detailed cluster information can be founded here: <https://hprc.tamu.edu/resources/>

Collaborative work on Graphcore Intelligence Processing Unit (IPU)



- Graphcore's IPU uses on-chip SRAM as a key component. This design allows for efficient and high-speed memory access.
- The IPU architecture is designed to minimize data movement between SRAM and off-chip memory, which helps reduce power consumption and latency, making it well-suited for AI and machine learning workloads.

Performance of Intelligence Processing Unit (IPU)



- Comparison of GPUs and IPUs on the ResNet 50 model in PyTorch with mixed precision.
- Scaling is similar for GPUs and IPUs, with similar magnitude per-device.

- While NVIDIA A100 series GPUs are indeed powerful, when compared to the Graphcore IPU, they still fall short by a considerable margin.
- The IPU has a linear scaling performance with the amount of IPU.

User case of Intelligence Processing Unit (IPU)

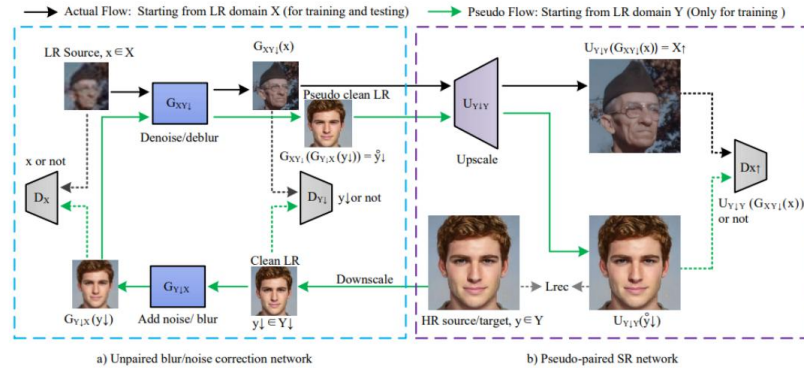


Figure: Architecture of Trans-CycleGAN for historical image super-resolution.

Table 2: Training performance of IPUs on the TransCycleGAN

Accelerator	Count	Images/sec
IPU	1	438
IPU	16	6552

- As a use case, we ported our image super-resolution model, TransCyCycleGAN, to the Graphcore IPU.
- Joint publication in the PEARC23: Porting AI/ML Models to Intelligence Processing Units (IPUs).

Lab 1: Logging to Launch Cluster & NVIDIA A30 GPUs

Log into Launch Using the HPRC Portal

HPRC webpage: <https://hprc.tamu.edu/>, Portal dropdown menu

The screenshot shows the Texas A&M High Performance Research Computing (HPRC) website. The header includes the Texas A&M logo and the text "TEXAS A&M HIGH PERFORMANCE RESEARCH COMPUTING". The navigation bar contains links: Home, User Services, Resources, Research, Policies, Events, Training, About, and Portal. The Portal dropdown menu is open, displaying the following options: Terra Portal, Grace Portal, FASTER Portal, FASTER Portal (ACCESS), ACES Portal (ACCESS), and Launch Portal (ACCESS). The Launch Portal (ACCESS) option is highlighted with a yellow box. The main content area features a simulation of a blue tractor-trailer impacting a yellow barrier system, labeled "MASH 5-12 - Tractor-Trailer Impacting Barrier System with 50 mi/h at 15 degrees". The simulation is titled "LS-DYNA keyword deck by LS-PrePost" and shows a time of 1.28. The Texas A&M Transportation Institute logo is also visible. On the left side, there are sections for "Quick Links" and "User Guides". The "Quick Links" section includes: New User Information, Accounts, Apply for Accounts, Manage Accounts, User Consulting, Training, Knowledge Base, Software, and FAQ. The "User Guides" section includes: Launch, ACES, FASTER, Grace, Terra, Portal, and Galaxy. At the bottom left, there is a "Cluster Status" link. The background of the main content area shows server racks.

ATM TEXAS A&M HIGH PERFORMANCE RESEARCH COMPUTING

Home User Services Resources Research Policies Events Training About Portal

Quick Links

- New User Information
- Accounts
- Apply for Accounts
- Manage Accounts
- User Consulting
- Training
- Knowledge Base
- Software
- FAQ

User Guides

- Launch
- ACES
- FASTER
- Grace
- Terra
- Portal
- Galaxy

Cluster Status

LS-DYNA keyword deck by LS-PrePost
Time = 1.28

Terra Portal

Grace Portal

FASTER Portal

FASTER Portal (ACCESS)

ACES Portal (ACCESS)

Launch Portal (ACCESS)

Texas A&M Transportation Institute

MASH 5-12 - Tractor-Trailer Impacting Barrier System with 50 mi/h at 15 degrees

Development and Testing of Structurally Independent Foundations for a 54" Tall High-Speed Containment Single Slope Concrete Barrier

Accessing Launch via the Launch Portal (ACCESS)

Log-in using your ACCESS credentials.

The screenshot shows the ACCESS login portal. At the top, there is a header with the ACCESS logo and "Powered By CILogon". Below the header is a "Consent to Attribute Release" section with a dropdown arrow. The consent text states: "TAMU FASTER ACCESS OOD requests access to the following information. If you do not approve this request, do not proceed." followed by a list of requested information: "Your CILogon user Identifier", "Your name", "Your email address", and "Your username and affiliation from your identity provider". Below the consent section is a "Select an Identity Provider" section. It features a dropdown menu with "ACCESS CI (XSEDE)" selected. There is a "Remember this selection" checkbox and a "Log On" button. A note below the button says "By selecting 'Log On', you agree to the privacy policy." At the bottom, there is a footer with links for questions, help, and support.

This is a close-up of the "Select an Identity Provider" dropdown menu. The dropdown is open, showing "ACCESS CI (XSEDE)" as the selected option. A yellow box highlights the dropdown menu.

Select the Identity Provider appropriate for your account.

The screenshot shows the ACCESS login portal. At the top, there is a header with the ACCESS logo and "Powered By CILogon". Below the header is a "Login to CILogon" section. It features a form with "ACCESS Username" and "ACCESS Password" fields, a "Log On" button, and a "Don't Remember Login" checkbox. To the right of the form is the CILogon logo and a description: "CILogon facilitates secure access to CyberInfrastructure (CI)." Below the logo, there is a warning icon and text: "If you had an XSEDE account, please enter your XSEDE username and password for ACCESS login". Below this, there are links for "Register for an ACCESS Account", "Forgot your password?", and "Need Help?". At the bottom, there is a link for "Click Here for Assistance".

It uses two-factor authentication with DUO.

Shell access via the HPRC Portal

–Top Banner Menu “Clusters” -> “Shell Access”

Launch OnDemand Portal Files ▾ Jobs ▾ Clusters ▾ Interactive Apps ▾ Dashboard ▾  My Interactive Sessions

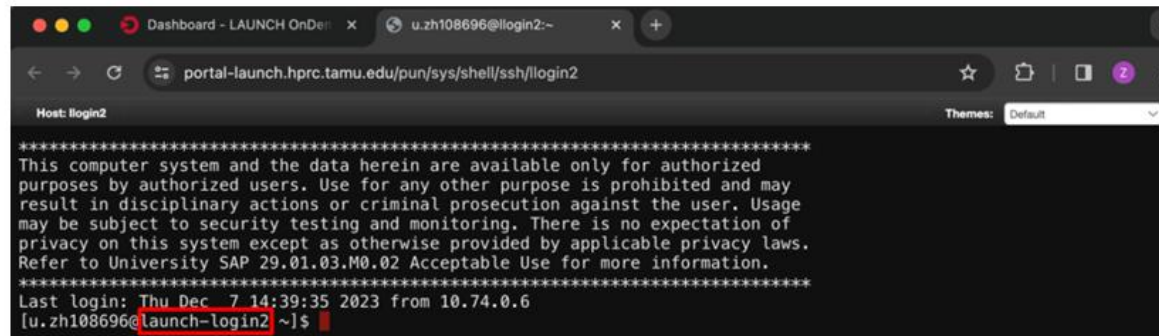
➤_launch Shell Access



OnDemand provides an integrated, single access point for all of your HPC resources.

Success!

- Welcome to the Launch login node!
- Check which login node you are on.



The screenshot shows a terminal window with a dark background. The title bar at the top indicates the connection is to 'u.zh108696@llogin2:~'. The address bar shows the URL 'portal-launch.hprc.tamu.edu/pun/sys/shell/ssh/llogin2'. The terminal content displays a standard SSH login banner with a disclaimer about system usage and privacy. Below the banner, it shows the last login time as 'Thu Dec 7 14:39:35 2023 from 10.74.0.6'. The prompt is '[u.zh108696@launch-login2 ~]\$', where 'launch-login2' is highlighted with a red box, indicating the user is on the correct login node.

```
Host: llogin2
*****
This computer system and the data herein are available only for authorized
purposes by authorized users. Use for any other purpose is prohibited and may
result in disciplinary actions or criminal prosecution against the user. Usage
may be subject to security testing and monitoring. There is no expectation of
privacy on this system except as otherwise provided by applicable privacy laws.
Refer to University SAP 29.01.03.M0.02 Acceptable Use for more information.
*****
Last login: Thu Dec 7 14:39:35 2023 from 10.74.0.6
[u.zh108696@launch-login2 ~]$
```

Access A30 GPUs

Check the A30 GPUs by using the command below.

```
$ pestat -p gpu -G
```

```
Or $ pestat -p gpu -G | grep a30
```

```
[u.lz104503@launch-login2 ~]$ pestat -p gpu -G
```

```
Print only nodes in partition gpu
```

```
GPU GRES (Generic Resource) is printed after each JobID
```

Hostname	Partition	Node	Num_CPU	CPUload	Memsize	Freemem	GRES/node	Joblist
		State	Use/Tot	(15min)	(MB)	(MB)		JobID User GRES/job ...
lg01	gpu	idle	0 192	0.00	760000	698405	gpu:a30:2(S:0-1)	
lg02	gpu	idle	0 192	0.00	760000	756214	gpu:a30:2(S:0-1)	
lg03	gpu	idle	0 192	0.00	760000	764855	gpu:a30:2(S:0-1)	
lg04	gpu	idle	0 192	0.00	760000	767164	gpu:a30:2(S:0-1)	
lg05	gpu	idle	0 192	0.00	760000	765299	gpu:a30:2(S:0-1)	
lg06	gpu	idle	0 192	0.00	760000	768472	gpu:a30:2(S:0-1)	
lg07	gpu	idle	0 192	0.00	760000	766180	gpu:a30:2(S:0-1)	
lg08	gpu	idle	0 192	0.00	760000	767975	gpu:a30:2(S:0-1)	
lg09	gpu	idle	0 192	0.00	760000	768860	gpu:a30:2(S:0-1)	
lg10	gpu	idle	0 192	0.00	760000	768878	gpu:a30:2(S:0-1)	

Lab 2: Run Two AI models on A30 GPUs

Commands to copy the materials

- Navigate to your personal scratch directory
`$ cd $SCRATCH`
- Clone the repository into your personal scratch directory
`$ git clone https://github.com/zhailujun123/A30.git`
- Enter this directory
`$ cd A30`
- 4 files are listed
`$ ls`
`PyTorch_A30.slurm PyTorch_MNIST.py TF_A30.slurm TF_MNIST.py`

Option 2

Launch OnDemand Portal

Files ▾

Jobs ▾

Clusters ▾

Interactive Apps ▾

Dashboard ▾

My Interactive Sessions

Home Directory

/scratch/user/u.lz104503

Open in Terminal

Refresh

+ New File

Home Directory

/scratch/user/u.lz104503



/ scratch / user / u.lz104503 / A30 /

Change directory

	Type	Name	Size
☐		PyTorch_A30.slurm	822 Bytes
☐		PyTorch_MNIST.py	2.96 KB
☐		TF_A30.slurm	1.1 KB
☐		TF_MNIST.py	1.55 KB

Introduction to Slurm file

	Type	Name		Size
☐		PyTorch_A30.slurm		822 Bytes
☐		PyTorch_MNIST.py		2.96 KB
☐		TF_A30.slurm		1.1 KB
☐		TF_MNIST.py		55 KB

 View
 Edit
 Rename
 Download

 Delete

Introduction to Slurm file

```
#!/bin/bash
##NECESSARY JOB SPECIFICATIONS

#SBATCH --job-name=TF_MNIST
#SBATCH --time=01:00:00
#SBATCH --nodes=1
#SBATCH --output=TF_MNIST.slurm_run.%j
#SBATCH --mem=480GB
#SBATCH --partition=gpu
#SBATCH --gres=gpu:a30:1
##SBATCH --odelist=lg01
##SBATCH --exclusive
##SBATCH --reservation=training

# Load required modules in the correct order
module load WebProxy
#module load GCCcore/11.3.0 Python/3.10.8
module load GCCcore/12.2.0 Python/3.10.8
module load GCC/12.2.0 OpenMPI/4.1.4 CUDA/11.7.0 TensorFlow/2.13.0

# Verify module loading
module list

# Check Python version
#python_version=$(python --version 2>&1)
#echo "Python Version: $python_version"

# Activate the Python virtual environment
python -m venv cvenv # Create virtual env
source cvenv/bin/activate

# Upgrade pip and resolve TensorFlow dependencies
pip install --upgrade pip
pip install wheel==0.36.2 six==1.16.0 requests==2.25.1 packaging==20.9 typing-extensions==3.7.4.3 urllib3==1.26.5

cd $SCRATCH/A30
python TF_MNIST.py
```

Specifies the maximum runtime for the job.

Multiple nodes can be requested

Job submission—Run a Tensorflow model on A30

- Use Sbatch command to submit a job

\$ sbatch TF_A30.slurm

```
[u.lz104503@launch-login2 A30]$ sbatch TF_A30.slurm  
Submitted batch job 1819  
[u.lz104503@launch-login2 A30]$
```

Job status-Two ways to check

- Check job status

\$ pestat -p gpu -G

```
[u.lz104503@launch-login2 A30]$ pestat -p gpu -G
Print only nodes in partition gpu
GPU GRES (Generic Resource) is printed after each JobID
Hostname Partition Node Num_CPU CPUload Memsize Freemem GRES/node Joblist
State Use/Tot (15min) (MB) (MB) JobID User GRES/job ...
lg01 gpu mix 1 192 0.11* 760000 698373 gpu:a30:2(S:0-1) 1824 u.lz104503 gpu:a30=1
lg02 gpu idle 0 192 0.06 760000 756201 gpu:a30:2(S:0-1)
lg03 gpu idle 0 192 0.11 760000 764508 gpu:a30:2(S:0-1)
lg04 gpu idle 0 192 0.00 760000 767161 gpu:a30:2(S:0-1)
lg05 gpu idle 0 192 0.00 760000 765299 gpu:a30:2(S:0-1)
lg06 gpu idle 0 192 0.00 760000 768471 gpu:a30:2(S:0-1)
lg07 gpu idle 0 192 0.00 760000 766179 gpu:a30:2(S:0-1)
lg08 gpu idle 0 192 0.00 760000 767974 gpu:a30:2(S:0-1)
lg09 gpu idle 0 192 0.00 760000 768860 gpu:a30:2(S:0-1)
lg10 gpu idle 0 192 0.00 760000 768877 gpu:a30:2(S:0-1)
```

Launch OnDemand Portal Files ▾ Jobs ▾ Clusters ▾ Interactive Apps ▾ Dashboard ▾ My Interactive Sessions Help ▾

Active Jobs

Show 50 entries

Active Jobs

- Active Jobs
- Drona Composer
- Drona Joblisting
- Job Composer

ID	Name	User	Account	Time Used	Queue	Status
1823	TF_MNIST	u.lz104503		00:01:01	gpu	Completed

Job submission—Run a Pytorch model on A30

- Use Sbatch command to submit a job
\$ sbatch PyTorch_A30.slurm

Thank you !!

Vim - a text editor

```
vim your_file.py
```

1. ``vim your_file.py``
 - This command opens the file named ``your_file.py`` in the Vim text editor. If the file does not exist, Vim will create a new file with that name.
2. ``I``
 - This command is used in normal mode in Vim to switch to insert mode before the cursor. When you press ``I``, Vim will allow you to start typing text at the current cursor position.
3. ``ESC``
 - The ``Esc`` key is used to exit from insert mode or any other mode and return to normal mode. It's a crucial key in Vim for transitioning between different modes.
4. ``:w`, `:wq`, `:q!``
 - These are command-line commands you can use while in normal mode or command-line mode in Vim.
 - ``:w``: Writes (saves) the changes to the file without quitting Vim.
 - ``:wq``: Writes the changes to the file and quits Vim.
 - ``:q!``: Quits Vim without saving changes, discarding any modifications made.